



KILLARA HIGH SCHOOL
YEAR 12 TERM 2 MATHEMATICS EXAMINATION
JUNE 2009
TOTAL TIME: 70 MINUTES
TOTAL MARKS: 66

DIRECTIONS TO CANDIDATES

- Attempt all questions
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved Calculators may be used.
- Answer each question in a SEPARATE Writing Booklet.
- You may ask for extra Writing Booklets if you need them.

QUESTION 1**Total marks: 30****EXPONENTIAL AND LOGARITHMIC FUNCTIONS**

- (a) Evaluate $e^{\log_e(2x+3)}$ when $x = 3$, correct to 2 significant figures. (2)
- (b) If $\log_m 2 = 0.623$ and $\log_m 3 = 0.847$, evaluate $\log_m 12$. (2)
- (c) Find $f''(5)$ if $f(x) = \log_e 2x$. (2)
- (d) Differentiate the following,.
- (i) $\frac{d}{dx}(\log_e 6x)$ (1)
- (ii) $\frac{d}{dx}(2x^3 - 4e^{3x})$ (1)
- (iii) $\frac{d}{dx} \log_e \left(\frac{x+1}{x-1} \right)$ (2)
- (iv) $\frac{d}{dx} e^x \log_e (x^2 - 2)$ (2)
- (v) $\frac{d}{dx} \log \sqrt[3]{5x}$ (2)
- (e) Find the Indefinite Integrate of the following with respect to x.
- (i) $\int 4e^x dx$ (1)
- (ii) $\int \frac{3}{2x} dx$ (1)
- (iii) $\int \frac{6}{e^{2x}} dx$ (1)
- (iv) $\int \frac{x}{3x^2 - 2} dx$ (1)
- (f) (i) Using the Trapezoidal Rule with three function values, calculate the area between the curve $y = \log_e x$, the x axis and the lines $x = 1$ and $x = 3$. (2)
- (ii) Find the exact volume of revolution for the region bounded by the curve $y = e^{2x}$, the x-axis and the lines $x = 2$ and $x = 3$ when it is rotated about the x-axis. (2)
- (g) (i) Show there is a minimum when $f(x) = e^{2x} - 2e^x - 12x$. (2)
- (ii) What is this minimum value, correct to 3 decimal places (2)
- (h) The gradient of the tangent at any point on a curve is given by $\frac{dy}{dx} = e^{-2x}$.
- (i) If the curve passes through the point $(0, 1)$, find the equation of the curve. (2)
- (ii) Find the point on the curve that has an abscissa of $x = -\frac{1}{2} \log_e 3$. (2)

(1)

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$$1a) \frac{\log_e(2x+3) \log_e 9}{e^{\log_e(2x+3) \log_e 9}} = e^9$$

$$b) \log_m 12 = \log_m 3 \times \log_m 4$$

$$= \log_m 3 \times \log_m 2^2$$

$$= 0.847 \times 2 \times 0.623$$

$$= 1.06$$

$$c) f(x) = \log_e 2x$$

$$f'(x) = \frac{2}{2x} = \frac{1}{x} = x^{-1}$$

$$f''(x) = -1 \times x^{-2}$$

$$= -\frac{1}{x^2}$$

$$f''(5) = -\frac{1}{25}$$

$$di) \frac{d}{dx} \log_e 6x = \frac{6}{6x}$$

$$= \frac{1}{x}$$

$$ii) \frac{d}{dx} (2x^3 - 4e^{3x}) = 6x^2 - 4e^{3x} \cdot 3$$

$$= 6x^2 - 12e^{3x}$$

$$= 6(x^2 - 2e^{3x})$$

$$iii) \frac{d}{dx} \log_e \frac{x+1}{x-1}$$

$$u = \frac{x+1}{x-1} \quad u' = \frac{x-1 - (x+1)}{(x-1)^2}$$

$$diii) \text{ continue } u' = -\frac{2}{(x-1)^2}$$

$$\frac{d}{dx} \log_e u = \frac{du}{u} \times \frac{du}{dx}$$

$$= \frac{1}{(x+1)} \times \frac{2}{(x-1)^2}$$

$$= \frac{(x-1)}{(x+1)} \times \frac{-2}{(x-1)^2}$$

$$= -\frac{2}{x^2-1}$$

$$iv) \frac{d}{dx} e^x \log_e (x^2-2) =$$

$$= e^x \cdot \log_e (x^2-2) + e^x \cdot \frac{2x}{x^2-2}$$

$$= e^x \left[\log_e (x^2-2) + \frac{2x}{x^2-2} \right]$$

$$v) \frac{d}{dx} \log_e \sqrt[3]{5x} = \frac{d}{dx} \log_e (5x)^{\frac{1}{3}}$$

$$= \frac{1}{(5x)^{\frac{1}{3}}} \times \frac{1}{3} (5x)^{-\frac{2}{3}} \times 5$$

$$= \frac{5}{3 (5x)^{\frac{1}{3}} (5x)^{\frac{2}{3}}}$$

$$= \frac{5}{3 (5x)}$$

$$= \frac{1}{3x}$$

(2)

$$e) i) \int 4e^x dx = 4e^x + C$$

$$ii) \int \frac{3}{2x} dx = \frac{3}{2} \ln x + C$$

$$iii) \int \frac{6}{e^{2x}} dx = 6 \int e^{-2x} dx$$

$$= 6 \frac{e^{-2x}}{-2} + C$$

$$= -3e^{-2x} + C \text{ or } -\frac{3}{e^{2x}} + C$$

$$iv) \int \frac{2}{3x^2-2} dx = \frac{1}{6} \int \frac{6x}{3x^2-2} dx$$

$$= \frac{1}{6} \ln(3x^2-2) + C$$

$$f) \text{ graph of } y = \ln x \text{ from } x=1 \text{ to } x=3$$

$$h = \frac{3-1}{2} = 1$$

$$i) A = \frac{h}{2} [f(1) + 2f(2) + f(3)]$$

$$= \frac{1}{2} [\log_e 1 + 2\log_e 2 + \log_e 3]$$

$$A = 1.24 \text{ unit}^2$$

$$ii) y = e^{2x}$$

$$V = \pi \int_2^3 y^2 dx \quad y^2 = e^{4x}$$

$$= \pi \int_2^3 e^{4x} dx$$

$$= \left[\pi \frac{e^{4x}}{4} \right]_2^3$$

$$= \frac{\pi}{4} [e^{12} - e^8] = \frac{\pi e^8}{4} (e^4 - 1)$$

$$g) i) y = e^{2x} - 2e^x - 12x$$

$$\frac{dy}{dx} = e^{2x} \cdot 2 - 2e^x - 12$$

$$\frac{dy}{dx} = 0 \text{ when there is max/min}$$

$$e^{2x} \cdot 2 - 2e^x - 12 = 0$$

$$e^{2x} - e^x - 6 = 0 \text{ let } u = e^x$$

$$u^2 - u - 6 = 0$$

$$(u-3)(u+2) = 0$$

$$u-3=0 \quad u+2=0$$

$$u=3 \quad u=-2$$

$$e^x=3 \quad e^x=-2$$

$$\ln e^x = \ln 3 \quad \log_e e^x = \log_e (-2)$$

$$x = \ln 3 \quad \text{no solution}$$

Stationary point at $x = \ln 3$

Nature of stationary point:

$$\frac{d^2y}{dx^2} = 2 \cdot e^{2x} \cdot 2 - 2e^x$$

$$= 4e^{2x} - 2e^x$$

$$\text{for } x = \ln 3 \quad y'' = 4e^{2\ln 3} - 2e^{\ln 3}$$

$$= 4e^{\ln 3^2} - 2e^{\ln 3}$$

$$= 4 \times 9 - 2 \times 3$$

$$= 30$$

$$h) i) \frac{dy}{dx} = e^{-2x} \quad (0,1)$$

(3)

h i continue

$$\frac{dy}{dx} = e^{-2x}$$

$$\int e^{-2x} dx = \frac{e^{-2x}}{-2} + C$$

$$y = -\frac{e^{-2x}}{2} + C$$

At (0, 1)

$$1 = -\frac{e^0}{2} + C$$

$$1 = -\frac{1}{2} + C$$

$$C = \frac{3}{2}$$

$$y = -\frac{e^{-2x}}{2} + \frac{3}{2} \text{ equation of the curve}$$

$$\text{ii)} \quad x = \frac{1}{2} \log_e 3$$

$$y = -\frac{e^{-2(\frac{1}{2} \log_e 3)}}{2} + \frac{3}{2}$$

$$= -\frac{e^{-\log_e 3}}{2} + \frac{3}{2}$$

$$= -\frac{e^{\log_e 3^{-1}}}{2} + \frac{3}{2}$$

$$= -\frac{3^{-1}}{2} + \frac{3}{2}$$

$$= -\frac{1}{3 \times 2} + \frac{3}{2} \Rightarrow y = -\frac{1}{6} + \frac{3}{2} = \frac{-1+9}{6} = \frac{8}{6} = \frac{4}{3}$$

$$\therefore \left(\frac{1}{2} \log_e 3, \frac{4}{3} \right)$$